

Analyzing Security Threats and Vulnerabilities in IoT Systems within Smart Cities

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Abstract—With the rapid advancement of technology and the increasing adoption of IoT devices in smart cities, it has become crucial to understand and address the potential risks associated with these systems. This study aims to analyze the security threats and vulnerabilities inherent in the interconnected Internet of Things (IoT) systems deployed within smart cities. Using a hybrid research approach, this paper integrates quantitative and qualitative methodologies to thoroughly analyze and identify prevailing security challenges in IoT frameworks in smart city environments. The result of the study highlights two types of security challenges security threats and security vulnerabilities along with their impact level. The most critical ones were Weak Authentication, Insecure Interfaces, Social Engineering Exploits, and Supply Chain Risks. This research contributes significantly to creating safer and more resilient smart cities by comparing various cases and synthesizing recommendations from the analyzed data. Furthermore, this investigation aligns with Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) and Sustainable Development Goal 11 (Sustainable Cities and Communities), thereby underscoring its relevance and urgency in the current global context.

Keywords—Analyzing, Security, Threats, Vulnerabilities, IoT (Internet of Things), Smart Cities, Safety.

I. INTRODUCTION

As urban areas worldwide are facing challenges towards improving efficiency, sustainability, and quality of life, in recent years, the concept of smart cities has gained significant attention [1]. One of the key components of smart cities is the interconnected Internet of Things (IoT) systems, which enable various devices and sensors to communicate and share data [2]. According to figure 1 [3], it is expected that by 2030, there will be over 29 billion IoT devices globally. Smart cities integrate a range of components to enhance their best functionality and innovation while minimizing possible security challenges. Among the components of smart cities for example, smart transport improves transportation systems but also needs to implement cybersecurity measures to prevent hacking of autonomous vehicles and ensure safe traffic management. In smart energy strategies focusing on optimizing energy consumption, it also needs to safeguard smart grids from cyber threats to maintain uninterrupted energy supply. The smart

technology developed for connectivity and data-driven decision-making, require robust cybersecurity protocols to protect sensitive information and infrastructure from cyber-attacks. For the sake of residents' well-being, smart living initiatives need to secure smart homes and healthcare data from unauthorized access. In smart education using through digital literacy, it is equally important to protect citizen against online threats. Finally, smart economy, safe city, and smart government initiatives to safeguard surveillance systems, drive economic growth, and enhance governance and transparency must ensure data privacy and protection against cyber threats and involve cybersecurity measures to ensure a secure and resilient urban environment overall.

While these components offer immense potential for advancement, they also introduce complex security challenges. With these systems' increasing in complexity and scale, there is a growing concern about the security threats and vulnerabilities they may face. Indeed, the interconnected IoT systems within smart cities are susceptible to a range of security threats and vulnerabilities in terms of confidentiality, integrity, and availability of the data and services provided by these systems. Some common security threats include the impersonation of legitimate users denying actions performed in the system and so on [4]. In addition to security threats, interconnected IoT systems within smart cities also face various vulnerabilities that can be exploited by attackers such as Data Breaches and Distributed Denial of Service (DDoS) [5].

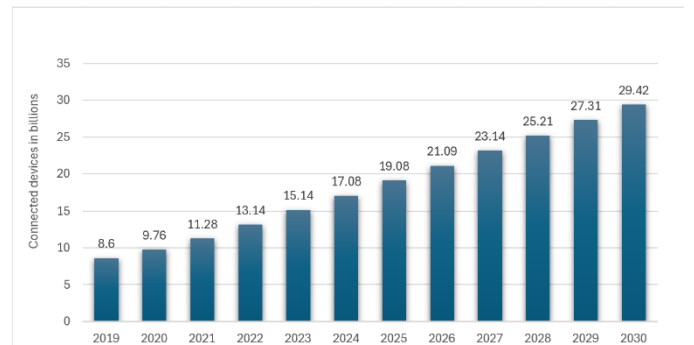


Fig. 1. Number of Internet of Things (IoT) connected devices worldwide from 2019 to 2023, with forecasts from 2022 to 2030.

The present study addresses the security threats and vulnerabilities in interconnected internet of things systems within smart cities. The goal is to analyze the security threats and vulnerabilities in IoT within smart cities. The primary objective is first to analyze the relevant literature review, analyze the different types of security threats and vulnerabilities in IoT in smart cities, including gaps, and then suggest some recommendations based on the findings. The significance of this study relies on its importance to society in general. The study is related to SDG #9 “Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation,” and SDG #11 “Make cities and human settlements inclusive, safe, resilient, and sustainable” [6].

The paper is structured as follows: Section 2 provides a comprehensive literature review about security threats and vulnerabilities in IoT in smart cities. Section 3 explains the methodology used in this study. Section 4 presents the findings of our study and discusses these findings within the context of IoT in smart cities and its security threats and vulnerabilities. Finally, Section 5 presents a conclusion and recommendations for future work.

II. LITERATURE REVIEW

The search strategy for the literature review of the study is articulated as follows: the review begins with an introductory context setting, which outlines the objectives and defines the scope. The methodology for identifying relevant literature involves searching through databases such as Google Scholar, lens.org, IEEE, and others. Search terms are combined strategically and employed with specific inclusion and exclusion criteria to refine the search results, particularly focusing on articles published within the last 10 years only. This process is followed by a synthesis of the existing literature and a critical evaluation of the solutions, approaches, or theories discovered, and assessing their effectiveness. Finally, the review concludes with the identification of gaps in existing knowledge, pinpointing any inconsistencies or areas for further research within the existing body of knowledge on the topic.

In a recent search we conducted on March 8th, 2024 using the Lens database (<https://www.lens.org/>) we found that since 2011, there have been a total of 1,514 papers published that analyze security threats and vulnerabilities in IoT systems, as depicted in Figure 2. However, only 445 of these papers have focused specifically on analyzing security threats and vulnerabilities within IoT in smart cities.

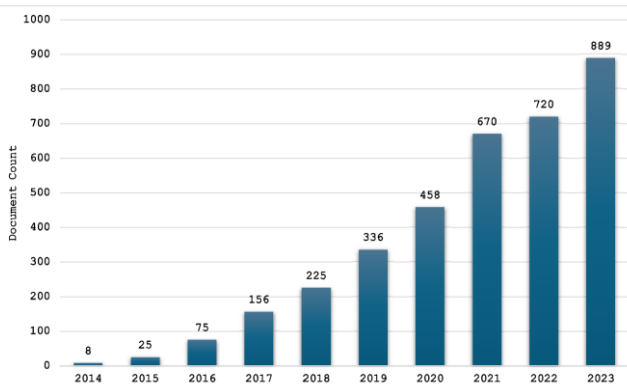


Fig. 2. Distribution of Papers Analyzing Security Threats and Vulnerabilities in IoT in Smart Cities using lens.org.

Data Breaches and Distributed Denial of Service (DDoS) attacks are examples of IoT vulnerabilities that can be exploited by attackers to gain unauthorized access which can lead to significant damage, disrupt services, and put lives at risk [5]. As the use of IoT increases in smart cities the risk of devices vulnerabilities and threats has become more prevalent [7]. Vulnerabilities are the weaknesses or flaws in IoT devices that could be exploited by attackers to gain unauthorized access or control [8]. On the other hand, threats refer to potential actions that could exploit vulnerabilities to cause harm or disruption such as malicious cyber attackers, and can lead to data breaches or physical damage to the devices.

References [5], [9] highlighted the concern of Distributed Denial of Service (DDoS) in IoT devices within smart cities. DDoS is when multiple computers flood a website or network with traffic, overwhelming it and causing it to become inaccessible to users. A primary concern of DDoS attacks is that they can cause significant disruption to critical infrastructure systems. Another concern associated with DDoS attacks involves the utilization of IoT devices as part of a botnet [5]. Malicious actors can compromise a huge number of IoT devices and use them as part of a botnet to launch DDoS attacks against targeted systems. It is crucial to raise awareness about the importance of IoT device security and educate users on how to protect their devices from potential vulnerabilities. By adopting a proactive approach to IoT device security, a safer and more secure IoT ecosystem can be established as reported by Neshenko et al. [10].

There is greater potential for malware to arise and inflict harm as IoT devices are used more frequently in smart cities. Malware is malicious software designed to disrupt, damage, or gain unauthorized access to devices systems according to [5]. This is particularly alarming considering the increasing frequency of IoT malware attacks across various sectors. For instance, according to Figure 3 [11], in 2022, the financial sector experienced a staggering year-over-year increase in IoT malware attacks. Then the retail industry and so on as shown in figure 3. Almost all of these sectors are part of the smart cities systems, thus, it is essential for smart city projects to prioritize the security of their cloud systems and work with cloud providers who have robust security measures in place to protect against malware attacks. By doing so, smart city projects can guarantee that they are able to operate safely and securely, without the risk of data loss or damage. [5]

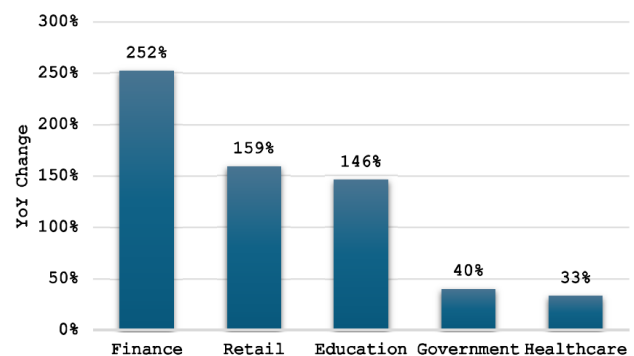


Fig. 3. Increase of customers targeted by IoT malware 2022, by sector.

In a study by Anwar et al. [4] 36 distinct threats through threat modelling were reported, they were categorized into four main components: sensing devices, communication channels, API/computation, and databases. The STRIDE method [12], created by Microsoft offers a way to analyze and tackle security risks in a system. In the context of IoT security within smart cities, several key threats have been identified: Impersonation (Spoofing) involves attackers masquerading as legitimate users to gain unauthorized access to systems or data, as discussed in reference [4]. Data Tampering refers to the alteration of legitimate information, undermining data integrity and reliability, also noted in reference [4]. Information Disclosure encompasses the unauthorized exposure of confidential information, leading to privacy breaches and data leakage, highlighted in reference [9]. Denial of Service attacks disrupt services, rendering systems unavailable to legitimate users and resulting in economic losses and user inconvenience, as detailed in references [4] and [9]. Lastly, Elevation of Privilege occurs when attackers acquire greater system privileges than originally granted, posing serious risks to system integrity and security, a concern raised in reference [5].

The exploration of security threats and vulnerabilities in IoT systems within smart cities, as detailed in the preceding sections, underscores the critical nature of this research area. As revealed through our comprehensive review, including the recent search yielding 1,514 papers related to IoT security, with only 445 focusing on smart cities, there are evident gaps in the current research landscape. These gaps are summarized in Table including an outline of their implications for future research, emphasizing the need for a more focused investigation into the unique security challenges of smart city contexts. The identified gaps such as the limited focus on smart cities, the need for a comprehensive threat analysis, and evaluation and integration of security measures underscore the significant need for an in-depth examination of these areas.

Table 1 shows important areas where more research is needed about security in IoT systems used in smart cities. It lists these areas and explains why they are important for future studies. This table helps us see that we need to understand and solve security problems in smart cities better.

TABLE 1 GAPS IN SECURITY THREATS AND VULNERABILITIES IN IoT WITHIN SMART CITIES

Identified Gap	Description	Future Research Needs
Focus on Smart Cities	Few studies specifically target smart cities.	Increase research dedicated to smart city IoT security.
Threat Analysis	Limited detail on evolving threats.	Analyze new and changing threats in smart cities.
Strategy Evaluation	Few evaluations of how well security measures work.	Test and validate security strategies in smart city contexts.
Impact Studies	Lack of data on the frequency and effects of threats.	Research the occurrence and impact of threats in smart cities.
Security Integration	Studies often look at security measures in isolation.	Explore how different security solutions work together in smart cities.

Human Factors	Not enough focus on human behavior's role in security.	Study how human behavior affects IoT security in smart cities.
Policy and Strategy	Need more guidance on managing security and privacy.	Develop and assess policies for better security in smart cities.

III. RESEARCH METHADODOLOGY

This study adopts a narrative approach to explore the security threats and vulnerabilities of IoT within smart cities. Data for this study were collected through a comprehensive literature review. To achieve this, different databases were searched systematically using Google Scholar, lens.org, IEEE and others in combination with various search terms related to security threats and vulnerabilities in IoT in smart cities, such as Security, Threats, Vulnerabilities, IoT (Internet of Things), Smart Cities, and Safety. The inclusion and exclusion criteria were applied to refine the search results by concentrating on articles published within the last decade and peer-reviewed articles. The synthesis of existing literature as well as critical evaluation of approaches, solutions or theories were done with the aim of identifying security challenges, their risks and possible solutions.

Through this review, we anticipate offering valuable insights into IoT security threats and vulnerabilities in smart cities. Despite the robustness of our approach, however, we acknowledge some limitations, such as the availability and accessibility of literature, the currency of information, and scope and coverage.

Ethical considerations were paramount throughout the study. Since the study primarily involved the analysis of existing literature, no direct involvement of human subjects was necessary, but proper citation and attribution were maintained to uphold academic integrity and respect intellectual property rights.

IV. RESULTS AND DISCUSSION

Our aim was to analyze the security threats and vulnerabilities inherent in the interconnected IoT systems deployed within smart cities. We reviewed and analyzed different types of papers with different objectives to reach our aim. Our results showed that we mainly have two types of security concerns of IoT within smart cities security threats and security vulnerabilities. This section of result and discussion will present the main results we found based on reviewing different types of articles from Google Scholar, lens.org, IEEE and others.

Table 2 outlines the various types of security vulnerabilities within different categories including network security, data security device security authentication and access control physical security, and supply chain security. Each category represents critical aspects of IoT systems that are susceptible to various security threats and weaknesses. A comprehensive overview of these security vulnerabilities is presented, highlighting potential weaknesses within IoT systems that need to be addressed through robust security measures, regular security assessments, firmware updates, secure authentication

mechanisms, encryption protocols, and supply chain security practices to reduce and eliminate the risks of IoT cyber threats.

TABLE 2 TYPES OF SECURITY VULNERABILITIES IN IoT WITHIN SMART CITIES

Category	Vulnerability Type	Description	Refs
Network Security	Distributed Denial of Service (DDoS)	Overloading systems to disrupt services using numerous IoT devices.	[13], [14]
	Insecure Network Services	Unprotected communication channels facilitate data interception	
Data Security	Data Breaches	Unauthorized access and extraction of sensitive information.	[5]
	Inadequate Encryption	Poor encryption practices expose data to eavesdropping.	
Device Security	Firmware Vulnerabilities	Compromises due to outdated or unpatched firmware.	[18],[21]
	Insecure Interfaces	Exploitable weaknesses in user interfaces and APIs	
Authentication & Access Control	Weak Authentication	Insufficient verification processes allow unauthorized access.	[20]
	Device Management Flaws	Poor configuration and management exposing security gaps.	
Physical Security	Social Engineering Exploits	Direct tampering with or damage to IoT devices	[15], [16]
	Supply Chain Risks	Introduction of vulnerabilities through compromised components.	

Table 3 provides a comprehensive overview of the different types of security threats that IoT systems encounter in smart cities. By categorizing these threats based on their characteristics, the table highlights the complex security landscape that smart cities are facing and serves as a critical foundation for policymakers, urban planners, and cybersecurity professionals, to prioritize and address the security challenges characteristic in smart city development.

TABLE 3 TYPES OF SECURITY THREATS IN IoT WITHIN SMART CITIES

Category	Threat Type	Description	Refs
Network Security	Malware Attacks	Malicious software designed to disrupt or gain unauthorized access to IoT networks.	[5]
	Man-in-the-Middle Attacks	Interception of communication between IoT devices and networks for malicious purposes.	[17]
Data Security	Data Interception	Unauthorized access to sensitive data during transmission within IoT systems.	[17],[19]
	Data Tampering	Unauthorized alteration or modification of data stored in IoT devices or networks.	[18]
Device Security	Exploitation of Firmware Vulnerabilities	Leveraging vulnerabilities in device firmware to gain control or access.	[21]
	Unauthorized Device Access	Unauthorized devices gaining access to IoT networks or systems.	[19]

Authentication & Access Control	Credential Theft	Stealing or compromising user credentials for unauthorized access.	[20]
	Access Control Bypass	Circumventing access control mechanisms to gain unauthorized access.	[20]
Physical Security	Device Tampering	Physical manipulation or damage to IoT devices to disrupt functionality.	[19]

Table 4 shows a list of vulnerability and threat types as we found in the literature. The table includes the impact level along with example cases and references. The impact level includes:

- Critical impact which refers to issues that are showstoppers and require immediate attention.
- High impact which refers to issues that, if not fixed in time, may cause hardware or data loss, or have a significant negative impact on the system or services.
- Moderate impact which refers to issues that may cause some data loss or have a moderate negative impact on the system or services.

This table would give a comprehensive view of how each type of vulnerability manifests in smart cities, including the level of impact each vulnerability has, and real or hypothetical examples that illustrate the issues. This approach provides a detailed and data-driven perspective on the security challenges faced by IoT implementations in urban environments.

Table 4 identifies the criticality of different vulnerabilities according to the impact levels (categorizing them from critical to moderate) and it makes concrete examples on those implications (can be seen in urban environment). The significant risk factors are the weak authentication, the insecure network services, the inadequate encryption, the firmware vulnerabilities, and the social engineering exploits, undermining the security of networks and internet of things devices owned by smart cities.

One of the important weaknesses that are joined major to the internet of things is autonomous access where the hackers can enter the smart devices and confidential data. Furthermore, neither secure service for data transmission will be created in public Wi-Fi zones. As result, personal information can be intercepted and unprotected. Furthermore, the table manifest the implication of plugging firmware loopholes and device management inadequacies that might lead to device compromise and damage on the overall system vibrancy. Vulnerabilities to cyber-attacks require a comprehensive approach to cybersecurity which should include secure authentication mechanisms, state of the art network protocols, continual software updates, as well active malware detecting and suppressing systems. By understanding the weakness of IoT area and trying to reduce them, all the smart city stakeholders can increase the security and the stability of the IoT systems, so that the whole urban infrastructure can be trusted completely and protected from malicious operations on the data plane.

TABLE 4 ANALYSIS OF VULNERABILITIES IN IoT WITHIN SMART CITIES

Vulnerability/Threat Type	Impact Level	Example Cases	Refs
Weak Authentication	Critical	Unauthorized access to smart devices	[20]
Insecure Network Services	High	Data interception in public Wi-Fi areas	[14]
Inadequate Encryption	Moderate	Eavesdropping on unencrypted data transmissions	[22]
Firmware Vulnerabilities	High	Compromise of devices due to outdated firmware	[21]
Device Management Flaws	Moderate	Security gaps from misconfigured devices	[18],[21]
Insecure Interfaces	Critical	Unauthorized control through weak APIs	[18]
Social Engineering Exploits	Critical	Direct tampering with or damage to IoT devices	[15]
Supply Chain Risks	Critical	Introduction of vulnerabilities through compromised components	[16]
Malware Attacks	High	Installation of malicious software leading to data theft or system compromise	[5]
Distributed Denial of Service (DDoS)	High	Disruption of services through overwhelming requests or traffic	[13]
Data Tampering	Critical	Unauthorized alteration or modification of data stored in IoT devices or networks.	[18]

V. CONCLUSION

This study adopts a narrative approach to explore the security threats and vulnerabilities of IoT within smart cities. Data for this study were collected through a comprehensive literature review. To achieve this, different types of papers from Google Scholar, lens.org, IEEE and others were reviewed and analyzed. The study identified two primary security challenges, each with its respective impact level. Firstly, security threats such as malware attacks were found to have a high impact level. Secondly, security vulnerabilities, including Weak Authentication and Insecure Interfaces, were deemed critical, while Insecure Network Services and DDoS were classified as high impact, and Device Management Flaws and Inadequate Encryption as moderate impact. We acknowledge some limitations, such as the availability and accessibility of literature, the currency of information, and scope and coverage. Future work should focus on how the interaction of humans affect the overall security of IoT within smart cities. In practical terms, the results of this study emphasize the critical need for implementing robust mitigation measures against security threats and vulnerabilities, along with the adoption of strategies aimed at minimizing their impact. By proactively addressing

these security challenges, IoT systems within smart cities can significantly enhance their overall security posture.

ACKNOWLEDGEMENT

The authors gratefully acknowledge the support from the College of Engineering and the College of Architecture and design at Effat University, Jeddah, Saudi Arabia. The authors also acknowledge the valuable discussion, suggestions and support received from Dr Tayeb Brahimi Researcher of Energy Lab at Effat College of Engineering, at Effat University, Jeddah, Saudi Arabia

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